

LAB EXPERIMENTS

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Experiment – 21 :

```
import pandas as pd
```

```
# Create DataFrame
```

```
data = {  
    "Name": ["John", "ALICE", "Bob", "DAVID", "Emma"],  
    "Country": ["India", "USA", "Canada", "UK", "Australia"]  
}
```

```
df = pd.DataFrame(data)
```

```
print("Original DataFrame:")
```

```
print(df)
```

```
# Swap the case of the Name column
```

```
df["Name"] = df["Name"].str.swapcase()
```

```
print("\nAfter swapping cases:")
```

```
print(df)
```

```

PS C:\Users\Vedvyass M\Music\Nutri-vision> c:; cd 'c:\Users\Vedvyass M\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundle
1 ALICE          USA
2   Bob          Canada
3 DAVID          UK
4   Emma        Australia

After swapping cases:
   Name      Country
0  jOHN      India
1  alice      USA
2   bOB      Canada
3  david      UK
4  eMMA      Australia
PS C:\Users\Vedvyass M\Music\Nutri-vision>

```

Experiment – 22 :

```
import matplotlib.pyplot as plt
```

```
# Data
```

```
x = [1, 2, 3, 4, 5]
```

```
y = [2, 4, 6, 8, 10]
```

```
# Draw line
```

```
plt.plot(x, y)
```

```
# Label X-axis
```

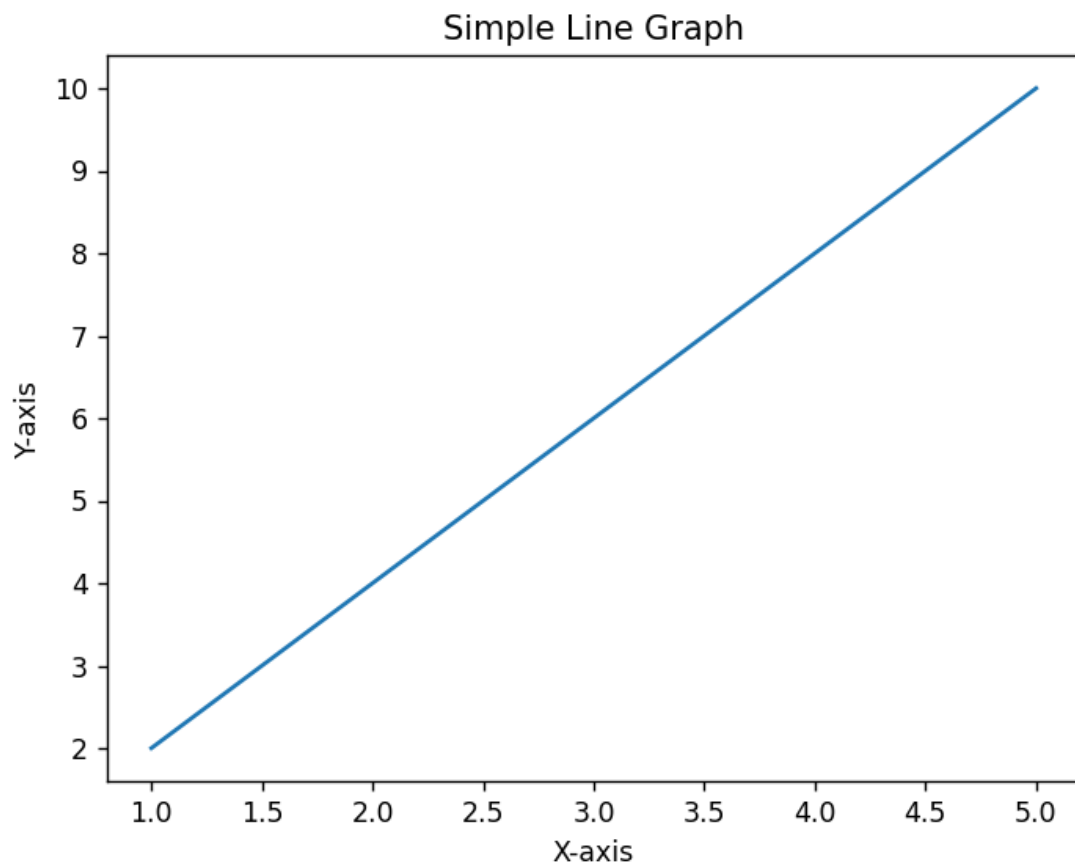
```
plt.xlabel("X-axis")
```

```
# Label Y-axis
```

```
plt.ylabel("Y-axis")
```

```
# Add title
plt.title("Simple Line Graph")

# Display graph
plt.show()
```



Experiment – 23 :

```
import matplotlib.pyplot as plt
```

```
# Read data from text file
```

```
x = []
```

```
y = []
```

```
with open(r"C:\Users\Vedvyass M\OneDrive\Desktop\test.txt", "r") as file:
```

```
    for line in file:
```

```
        values = line.split()
```

```
        x.append(float(values[0]))
```

```
        y.append(float(values[1]))
```

```
# Draw line
```

```
plt.plot(x, y)
```

```
# Add labels and title
```

```
plt.xlabel("X-axis")
```

```
plt.ylabel("Y-axis")
```

```
plt.title("Line Graph from Text File")
```

```
# Display graph
```

```
plt.show()
```

